

Appendix 1.3: COMPUTE TOKEN ($T_{compute}$ / PoUC / PoUCC)

Transceptor Global / Technical-Economic Regulation of the Compute Contour
*Document Identifier: SPEC-COMPUTE-2026-V3.4-ADAPTIVE | Legal Frame: Swiss DLT-Act / ADGM / MAS Compliant***System Invariant:** $T_{compute}$ represents the immutable, utility-bound accounting vector for physical silicon computation and human cognitive effort within Transceptor Global. Direct fiat purchase of compute power is hardware-blocked at the kernel level (`clearing_gateway.hpp`) to eliminate speculative capital manipulation.

1. Purpose, Utility Status, and Fiat-Inflow Ban

The Compute Token ($T_{compute}$) serves as the exclusive utility currency and settlement instrument for hardware-isolated compute cycles, AI model inference, and verifiable expert cognitive labor across the Transceptor Global ecosystem.

1.1. Functional Utility Status

Unlike speculative cryptocurrencies or unbacked utility tokens, $T_{compute}$ is strictly bounded by hardware execution metrics inside Trusted Execution Environments (TEE / Secure Enclaves). It quantifies the execution of useful floating-point operations (FLOPs_{useful}) and verified human cognitive input.

1.2. Kernel-Level Ban on Direct Fiat Inflow

To prevent plutocratic capture and speculative inflation of computational resources, direct fiat currency acquisition of $T_{compute}$ is strictly prohibited.

- **Microcode Enforcement:** The underlying kernel logic in `clearing_gateway.hpp` rejects any direct fiat-to-$T_{compute}$ minting transactions.
- **P2P Barter Gateway:** Access to $T_{compute}$ is restricted exclusively to peer-to-peer (P2P) value barter against the primary value vector T_{net} (or verified RWA-backed collateral).

- **Barter Fee & Clearing Split:** A standardized 1.50% fee is applied to all $T_{net} \to T_{compute}$ barter conversions. Collected fees are split according to the treasury protocol: 50% to system liquidity reserves, 30% to node operators, and 20% to the Founder's Development Fund.
- **Guaranteed Valuation Floor:** To maintain economic stability for compute providers and station operators, $T_{compute}$ maintains a guaranteed base valuation floor of **0.30 € per quantum** ($1\text{ quantum} \equiv 10^{15} \text{ FLOPs}$ or standardized TEE compute unit).

2. Two-Layer Consensus Model (PoUC & PoUCC)

Transceptor Global replaces energy-wasteful Proof-of-Work (PoW) and plutocratic Proof-of-Stake (PoS) with a dynamic two-layer consensus architecture: Proof of Useful Computation (PoUC) at the silicon hardware level and Proof of Useful Cognitive Computation (PoUCC) at the combined socio-technical layer.

2.1. The Silicon Layer (PoUC)

The physical hardware layer consists of GPU, NPU, and CPU clusters operating within strict TEE enclaves (Open Enclave SDK / Strict PRM). Raw compute cycles are validated using cryptographic TEE remote attestation quotes, ensuring that 100% of executed operations perform real-world semantic parsing, vector search, or zero-knowledge proof generation.

2.2. The Carbon Layer (PoUCC)

Recognizing that human expert judgment, scientific research, and civic governance cannot be reduced solely to machine FLOPs, the system integrates human cognitive work into the unified consensus model.

2.3. Unified Consensus Equation

The total verified compute weight (PoUCC_{total}) is calculated according to the deterministic formula:

$$\text{PoUCC}*{total} = *text{FLOPs}*{useful} + H_{live_effective} \cdot K_{comp} \cdot (I_{cp} + I_{sc})$$

Where:

- FLOPs_{useful} represents the volume of verified, non-redundant floating-point operations executed inside TEE enclaves.
- $H_{live_effective}$ is the capped, outcome-validated live expert norm-hours.

- K_{comp} is the domain complexity coefficient assigned to the specific task or research track.
- I_{cp} is the user's verified Competence Index from their Soulbound DID passport.
- I_{sc} is the Social Contribution Index reflecting verified public benefit.

3. Anti-Spoofing & Outcome-Based Capping

To prevent Goodhart's Law ("when a measure becomes a target, it ceases to be a good measure") and combat automated metric farming, $T_{compute}$ implementation enforces strict outcome-based validation.

3.1. Outcome-Based Hour Normalization

Raw time logging (H_{fact}) cannot directly generate compute tokens or reputation increments. Effective hours ($H_{live_effective}$) are strictly constrained by the benchmark ceiling and physical/social impact multipliers:

$$H_{live_effective} = \min(H_{fact}, H_{benchmark}) \cdot \Delta S_{useful}$$

Where:

- H_{fact} is the actual verified time spent during active live sessions.
- $H_{benchmark}$ is the maximum standard hour ceiling for the specific task type.
- $\Delta S_{useful} \in [0.0, 1.0]$ is the entropy reduction factor, measuring real-world infrastructure, physical, or academic outcome delivered over time.

3.2. Dual-Blind ZK Cross-Validation & Anti-Deepfake Controls

- **Liveness & Biometric Tethering:** Live expert sessions require continuous passive optical photogrammetry (corneal reflection micro-blink tracking, capillary pulse detection) via the Personal Satellite to prevent deepfake injection or video replay attacks.
- **Protocol of Internal Blindness:** Expert contributions are processed via non-interactive zero-knowledge proofs (ZK-SNARKs). Personal identities, physical locations, and raw biometric signatures are completely isolated from reviewers and network nodes.
- **Dynamic Quantum Jitter:** To prevent bot farms from reverse-engineering smart-contract triggers, crediting timers introduce randomized delay pulses ($T_{credit} = T_{trigger} + \text{random}(\Delta t_{min}, \Delta t_{max})$).

3.3. Automated Sanction Triggers

Any detected attempt to farm metrics using synthetic generative media, automated click-bots, or prompt injection triggers immediate automated sanctions:

- Immediate forfeiture of pending $T_{compute}$ credits.
- Cascade reputation collapse dropping the offender's electoral weight to the minimum threshold ($W_{user} \to 0.01$) via a discrete penalty ($\Delta I_{cp} = -4.99$).

4. Compute Resource Quantization & Allocation Split

To ensure deterministic operation across all nodes, platform hardware resources (GPU, NPU, CPU) are quantized and strictly partitioned into three specialized functional allocations.

Resource Segment	Allocated Capacity	Target Hardware
Agora AI-Oracle Engine	**65%**	GPU / NPU
ZK-Shield & DID Infrastructure	**30%**	GPU / CPU
MPC & Latency Management	**5%**	CPU

4.1. Network Heartbeat and Latency Invariants

- **Ping-Heartbeat Interval:** The Lightweight Core maintains a background heartbeat transmitting encrypted stability certificates every **100 milliseconds** with a maximum drift tolerance of **$\pm 1\text{ ms}$**.
- **Emergency Fail-Safe:** If network drift exceeds latency limits or a hardware breach occurs, the Zeroization protocol wipes master keys from CPU cache within **10 milliseconds**.

5. Emission Rules & Zero-Idling Policy

$T_{compute}$ emission is dynamically calibrated against actual computational demand, preventing inflationary over-issuance and energy waste.

5.1. Dynamic Emission Mechanics

Tokens of $T_{compute}$ are minted exclusively upon cryptographic verification of completed, useful work units inside TEE enclaves. Unbacked minting or pre-mining is architecturally impossible.

5.2. Strict Zero-Idling Policy

- **Prohibition of Wasteful Cycles:** Nodes are forbidden from running idle loops or dummy calculations to artificially claim compute status.
- **Dynamic Task Reallocation:** If allocated capacity in the ZK-Shield (30%) or MPC (5%) sectors remains unutilized, idle GPU/NPU cycles are automatically re-routed to the Agora Semantic AI-Oracle queue to process background civic proposals.
- **Thermal & Power Optimization:** TEE execution modules enforce strict sub-2.15 ms computation limits per task vector to eliminate thermal throttling and optimize power draw on client devices.

6. Hybrid Integration of BigTech Enterprise APIs

While Transceptor Global relies primarily on its sovereign, open-source TEE AI-Oracle network, complex enterprise workflows require hybrid access to external state-of-the-art commercial LLMs (OpenAI Enterprise, Google Gemini, xAI Grok).Security Mandate: All interaction with external BigTech API providers must route through hardware-isolated TEE proxy enclaves implementing Zero-Data Retention (ZDR) and multi-provider anonymization.

6.1. Architectural API Routing Rules

1. **Semantic Anonymization:** Raw enterprise or civic data is de-identified and converted into abstract vector representations inside local TEE enclaves prior to outbound transmission. External API providers never receive raw personal identifiable information (PII) or unencrypted budget specifications.
2. **Multi-Provider Edge Routing:** The system dynamically routes requests across available LLM providers based on real-time task capability, availability, and pricing, maintaining a routing overhead latency under **4 milliseconds**.
3. **Zero-Data Retention (ZDR) Enforcement:** Integration contracts with external API vendors strictly require Zero-Data Retention agreements, preventing vendor model training on Transceptor telemetry or user queries.

6.2. Cost-Price Procurement via $T_{compute}$

B2B partners, engineering bureaus, and AI agent developers acquire commercial API access through $T_{compute}$ at **pure wholesale cost-price** without intermediary markups. The system acts as a transparent, high-volume aggregator, passing raw infrastructure savings directly to ecosystem participants.